

Rajalakshmi Engineering College

Name: Tharun B P
Email: 241501230@rajalakshmi.edu.in
Roll no: 241501230
Phone: 8148499002
Branch: REC
Department: AI & ML - Section 1
Batch: 2028
Degree: B.E - AI & ML

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

```
import java.util.*;

class DotException extends Exception {
    public DotException(String message) {
        super(message);
    }
}

class AtTheRateException extends Exception {
    public AtTheRateException(String message) {
        super(message);
    }
}

class DomainException extends Exception {
    public DomainException(String message) {
        super(message);
    }
}

class EmailValidator {
    public void validateEmail(String email) throws DotException,
        AtTheRateException, DomainException {
        if (email == null || email.isEmpty()) {
            throw new DotException("Invalid Dot usage");
        }

        int atCount = email.length() - email.replace("@", "").length();
        if (atCount != 1 || email.startsWith("@") || email.endsWith("@") ||
            email.contains("@@")) {
            throw new AtTheRateException("Invalid @ usage");
        }
    }
}
```

```

        if (!email.contains(".") || email.endsWith(".") || email.startsWith(".") ||
            email.contains("..")) {
            throw new DotException("Invalid Dot usage");
        }

        int atIndex = email.indexOf('@');
        int lastDotIndex = email.lastIndexOf('.');
        if (lastDotIndex < atIndex || lastDotIndex == email.length() - 1) {
            throw new DotException("Invalid Dot usage");
        }

        String domain = email.substring(lastDotIndex + 1);
        if (!(domain.equals("com") || domain.equals("in") || domain.equals("net") ||
            domain.equals("biz"))) {
            throw new DomainException("Invalid Domain");
        }

        System.out.println("Valid email address");
    }
}

```

```

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String email = scanner.nextLine();
        EmailValidator validator = new EmailValidator();

        try {
            validator.validateEmail(email);
        } catch (DotException e) {
            System.out.println("DotException: " + e.getMessage());
            System.out.println("Invalid email address");
        } catch (AtTheRateException e) {
            System.out.println("AtTheRateException: " + e.getMessage());
            System.out.println("Invalid email address");
        } catch (DomainException e) {
            System.out.println("DomainException: " + e.getMessage());
            System.out.println("Invalid email address");
        }

        scanner.close();
    }
}

```

}
}

Status : Correct

Marks : 10/10